

incorrect, treat as e.c.f.)

- | | | | | |
|--------------|-------------|---|----------------|-----|
| 5 (a) | (i) | (induced) e.m.f proportional/equal to rate of change of flux (linkage)
<i>(allow 'induced voltage, induced p.d.)</i>
flux is cut as the disc moves
hence inducing an e.m.f | B1
M1
A0 | [2] |
| | (ii) | field in disc is not uniform/rate of cutting not same/speed of disc not same (over whole disc)
so different e.m.f.'s in different parts of disc
lead to eddy currents | B1
M1
A0 | [2] |
| (b) | | eddy currents dissipate thermal energy in disc
energy derived from oscillation of disc
energy of disc depends on amplitude of oscillations | B1
B1
B1 | [3] |

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL - OCT/NOV 2006	9702	04

6 (a) (i) peak voltage = $6\sqrt{2}$

C1